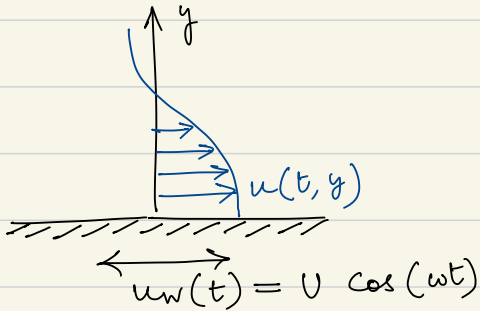


The Stokes second Problem



A flat plate, oscillates harmonically with circular frequency ω , bounding a semi-infinite fluid domain of dynamic viscosity μ .

We want to compute the total viscous dissipation rate per unit length of the plate

An aside on the viscous dissipation rate :

The viscous dissipation rate at x_i is given by $(\tau_{ij} \partial_i u_j)$

The velocity gradient tensor $\partial_i u_j = S_{ij} + \Omega_{ij}$

symmetric $\leftarrow$
 $\leftarrow$ anti-symmetric
viscous stress tensor

Hence, we have, $\tau_{ij} = 2\mu S_{ij}$ and $S_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i)$

Hence the dissipation rate per unit volume becomes,

$$\phi = 2\mu S_{ij} S_{ij} \quad \left(\because S_{ij} \Omega_{ij} = 0 \right)$$

The plate moves as, $u(t, y=0) = U \cos(\omega t)$, $\eta = \frac{\mu}{\rho}$

The solution is, $u(t, y) = U e^{-\sqrt{\frac{\omega}{2\nu}} y} \cos\left(\omega t - \sqrt{\frac{\omega}{2\nu}} y\right)$

$$= U e^{-y/\delta} \cos\left(\omega t - y/\delta\right)$$

$$= U e^{-ky} \cos\left(\omega t - ky\right); k = \frac{1}{\delta} = \sqrt{\frac{\omega}{2\nu}}$$

The only non-zero strain rate is $\frac{\partial u}{\partial y} \Rightarrow$

$$S_{ij} = \begin{bmatrix} \frac{1}{2} \left(\frac{\partial u}{\partial x}, \cancel{0} \right) & \frac{1}{2} \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \right) \\ \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial x} \right) & \frac{1}{2} \left(\frac{\partial v}{\partial y}, \cancel{0} \right) \end{bmatrix}$$

for the Stokes 2nd Problem, $v = 0$, $\frac{\partial v}{\partial x} = 0$, $\frac{\partial v}{\partial y} = 0$ and $\frac{\partial u}{\partial x} = 0$

$$\Rightarrow S_{ij} = \begin{bmatrix} 0 & \frac{1}{2} \frac{\partial u}{\partial y} \\ \frac{1}{2} \frac{\partial u}{\partial y} & 0 \end{bmatrix}$$

$$S_{ij} S_{ij} = \begin{bmatrix} 0 & \frac{1}{2} \frac{\partial u}{\partial y} \\ \frac{1}{2} \frac{\partial u}{\partial y} & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} \frac{\partial u}{\partial y} \\ \frac{1}{2} \frac{\partial u}{\partial y} & 0 \end{bmatrix} = S_{11} S_{11} + S_{12} S_{12} + S_{21} S_{21} + S_{22} S_{22}$$

$$= 0 + \frac{1}{4} \left(\frac{\partial u}{\partial y} \right)^2 + \frac{1}{4} \left(\frac{\partial u}{\partial y} \right)^2 + 0$$

$$= \frac{1}{2} \left(\frac{\partial u}{\partial y} \right)^2$$

$$\phi = \cancel{2} \mu \frac{1}{\cancel{2}} \left(\frac{\partial u}{\partial y} \right)^2 = \mu \left(\frac{\partial u}{\partial y} \right)^2$$

Take derivative,

$$\frac{\partial u}{\partial y} = U \left[-k e^{-ky} \cos(\omega t - ky) + e^{-ky} \sin(\omega t - ky) k \right]$$

$$= k U e^{-ky} \left[\sin(\omega t - ky) - \cos(\omega t - ky) \right]$$

$$\Rightarrow \phi = \mu k^2 U^2 e^{-2ky} \left[\sin(\omega t - ky) - \cos(\omega t - ky) \right]^2$$

$$= \mu k^2 U^2 e^{-2ky} \left[1 - 2 \sin(\omega t - ky) \cos(\omega t - ky) \right]$$

$$= \mu k^2 U^2 e^{-2ky} \left[1 - \sin(2\omega t - 2ky) \right] \rightarrow \text{dissipation}$$

Here $\omega = \frac{2\pi}{T}$

rate per unit volume.

Now, we integrate this over the time period $\int_t^{t+T} \phi dt$

$$= \int_t^{t+T} \mu k^2 U^2 e^{-2ky} \left[1 - \sin(2\omega t - 2ky) \right] dt$$

$$= \mu k^2 U^2 e^{-2ky} \left[t + \frac{\cos(2\omega t - 2ky)}{2\omega} \right]_t^{t+T}$$

$$= \mu k^2 U^2 e^{-2ky} \left[T + \frac{T}{4\pi} \cos(2\omega t + 2 \times 2\pi - 2ky) - \frac{T}{4\pi} \cos(2\omega t - 2ky) \right]$$

$$= \mu k^2 U^2 e^{-2ky} T = \bar{\phi}(t)$$

0 $\left(\because \cos(4\pi + \theta) = \cos(\theta) \right)$

This is also obvious earlier considering integral over $2 \sin(\) \cos(\) = 0$, over the period T .

Now, this can be integrated over the domain of length P ,

$$\Phi = \int_0^P \int_0^\infty \int_t^{t+T} \delta \, dt d\Omega = P \int_0^\infty \underbrace{\mu k^2 U^2 e^{-2ky} T}_{\mu \frac{\pi}{2\nu} U^2 e^{-2ky} \frac{2\pi}{\omega}} dy$$

$$\mu \frac{\pi}{2\nu} U^2 e^{-2ky} \frac{2\pi}{\omega}$$

$$\Phi = P \int_0^\infty \mu \frac{\pi}{\nu} U^2 e^{-2ky} dy$$

$$= P \mu \frac{\pi}{\nu} U^2 \left[\frac{e^{-2ky}}{-2k} \right]_0^\infty = P \mu \frac{\pi}{\nu} U^2 \frac{1}{-2k} \left[\underbrace{\frac{1}{e^\infty}}_0 - \underbrace{\frac{1}{e^0}}_1 \right]$$

$$= P \cancel{\delta} \frac{\pi}{\cancel{\nu}} U^2 \frac{1}{2} \sqrt{\frac{2\nu}{\omega}}$$

$$\Phi = \frac{P}{2} \delta \pi U^2 \sqrt{\frac{2\nu}{\omega}} = \frac{P}{2} \delta \pi U^2 \delta = \frac{P}{2} \delta \sqrt{\pi} U^2 \sqrt{\frac{\nu}{f}}$$

If we define a kinetic energy scale for an oscillating fluid volume $H \times W$ as $E_{KE} = H W \delta \frac{U^2}{2}$

and Womersley number is defined as $Wo = \frac{W}{\delta_{st}} = \frac{W}{\sqrt{\nu/\omega}}$

$$\text{We get } \frac{\Phi^*}{E_{KE}} = \frac{\Phi}{H} \frac{\sqrt{2}}{W/\delta_{st}} = \frac{\pi P \sqrt{2}}{H} \frac{1}{(Wo)}$$